PREDICTING PRODUCT YIELD USING MACHINE LEARNING

THE MACHINE LEARNING PIPELINE

Data Ingestion
Ensures clean and relevant data is available for processing.

Data Wrangling
Cleans and transforms data into a usable format.

Analysis and Modelling
Identifies patterns and builds predictive models.

Prediction and Optimisation
Uses models to make predictions and improve outcomes.

Reporting and Visualisation
Presents findings and insights effectively.

MODELLING STRATEGY

To find the best way to forecast the manufacturing yield, two different methods were tested - high-level batch aggregations against sequential time-series models (using lag features).

Batch Statistics

Product Mean

Time Series Models

Product_t = f(Xt, Xt-1, Xt-2)

Product Mean = Mean(Product_t)

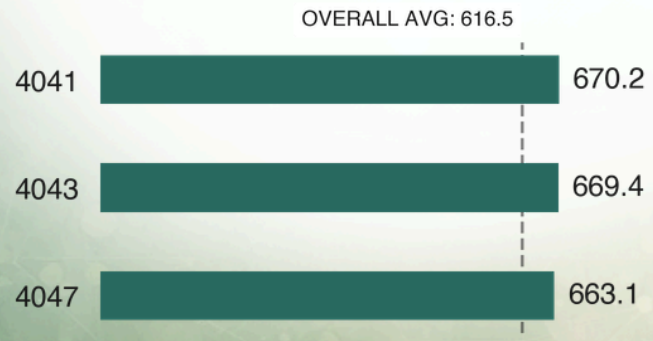
Product Rate = Product Mean * Mean Total Liquid Inflow Rate * 0.001

YIELD VOLATILITY

Batches 4041, 4043, and 4047 significantly outperformed the facility average of 616.5 kg/hr. This wide variance in production rates highlights that manufacturing is a highly volatile process, presenting a clear opportunity for process optimisation.

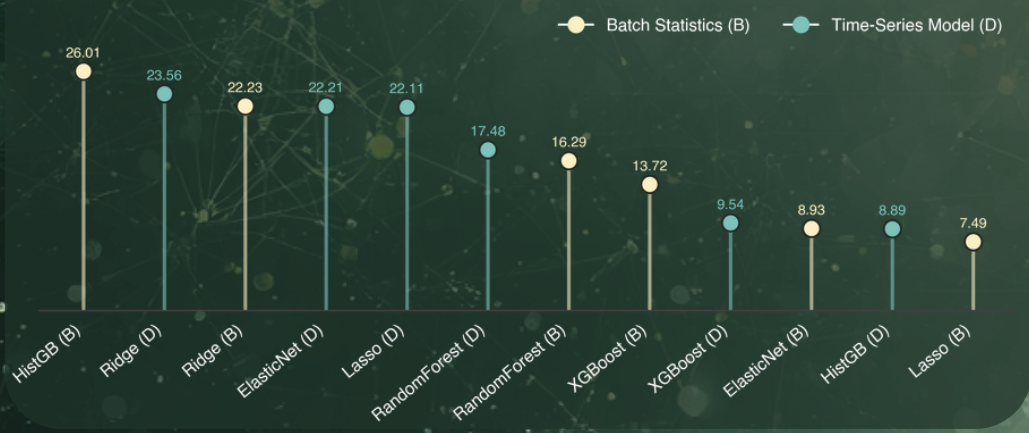
TOP 3 MOST PRODUCTIVE BATCHES

Highest product rates (kg/hr) of top 3 batches.



MODEL PERFORMANCE: RMSE RANKING

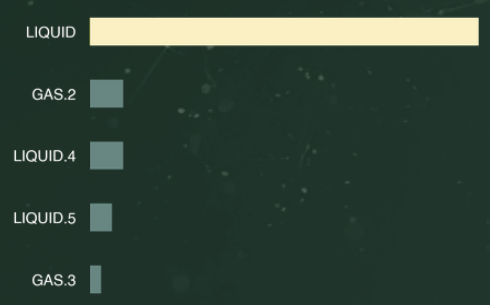
Comparing Batch Statistics vs. Time-Series Models (Lower error is better).



With a **RMSE** of just **7.49**, the Lasso Regression model utilizing Batch Statistics significantly outperformed the rest. It proved to be the most accurate and reliable method for forecasting our final manufacturing yields.

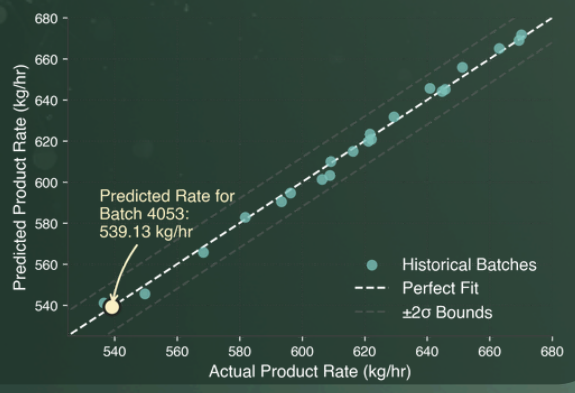
KEY YIELD DRIVERS

Process parameters most related to product rate for Lasso Model



FINAL MODEL INFERENCE

Actual vs. Predicted Yield across all training and test batches.



The Lasso Regression model identifies **LIQUID** as the main driver of product rate. Most predictions lie close to the perfect-fit line, with **Batch 4053** predicted at **539.13 kg/hr**.

Sources: Majumdar, S. (2025) 'Lag & Rolling Features: Time Series Forecasting with Python', Medium